



School of Future Tech

Case Study Report

on

Currency Exchange and Forex System using C++

by

Anjali Patil

150096725013

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1. Introduction to the Case Study

Introduction

Currency exchange is an essential part of international trade, travel, and global business. People often need to convert one country's currency into another based on exchange rates. A Forex (Foreign Exchange) system helps perform these conversions quickly and accurately.

The purpose of this case study is to develop a Forex Management System using the C++ programming language. The system allows users to convert currencies such as INR, USD, EUR, GBP, JPY, and AUD. It also stores transaction history in a file along with the date and time of each transaction.

This project demonstrates the use of programming concepts such as functions, file handling, conditional statements, loops, and menu-driven systems in C++.

2. Problem Statement / Case Background (Abstract)

Abstract

Currency conversion is frequently required in international financial transactions. However, manual conversion can be time-consuming and prone to calculation errors.

The objective of this project is to design and implement a Forex Management System that automates currency conversion. The program allows users to convert currencies using predefined exchange rates, view past transactions, and track the number of conversions performed in the session.

The system is developed using C++ and uses file handling to store transaction records for future reference.

3. Case Study Design

Case Study Design

The Forex Management System is designed as a menu-driven console application. The system allows users to perform currency conversions and manage transaction records.

The system provides multiple currency options such as INR, USD, EUR, GBP, JPY, and AUD. Users can select the source and destination currencies and enter the amount to convert. The system then calculates the converted value based on predefined exchange rates.

Each transaction is stored in a file named *transactions.txt* along with the date and time. Users can also view the transaction history and the total number of transactions performed during the session.

Main Functionalities

- Currency conversion
- Transaction history storage
- Display of total transactions
- Menu-driven user interface

4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

Technology Used

- C++ Programming Language
- File Handling (fstream)
- Time Library (ctime)
- Input/Output Library (iostream)
- Output Formatting (iomanip)

Programming Concepts Used

- Functions
- Conditional Statements
- Switch Case
- Loops
- File Handling
- Date and Time Handling

Algorithm

1. Start the program
2. Display the Forex Management System menu
3. User selects an option
4. If user selects currency conversion
 - Display available currencies
 - Take input for source currency
 - Take input for target currency
 - Take input amount
 - Apply exchange rate

- Display converted amount
- Save transaction in file
- 5. If user selects transaction history
 - Read transactions from file
 - Display them
- 6. If user selects total transactions
 - Display number of conversions
- 7. Repeat until user chooses Exit
- 8. End program

5. Problem Statement / Case Study Implementation Details and Snapshots.

Implementation

The Forex Management System was implemented using C++. The program uses multiple functions to organize the logic such as displaying available currencies, calculating exchange rates, converting currencies, and managing transaction history.

File handling is used to store transaction details in a text file. The system also records the date and time of each transaction using the time library.

Functions Used in the Program

- `showCurrencies()` – displays available currencies
- `getCurrencyName()` – returns the name of the currency
- `getRate()` – returns exchange rate between currencies
- `getDateTime()` – gets current date and time
- `convertCurrency()` – performs the currency conversion
- `viewTransactions()` – displays transaction history
- `showTotalTransactions()` – shows total transactions

EXPLORER

OPEN EDITORS

Welcome

main.cpp

CURRENCY EXCHANGE & ...

main

main.cpp

transactions.txt

OUTLINE

TIMELINE

Welcome

main.cpp

main()

1 #include <iostream>

2 #include <fstream>

3 #include <iomanip>

4 #include <ctime>

5 using namespace std;

6

7 int transactionCount = 0;

8

9 // Function to show currencies

10 void showCurrencies() {

11 cout << "\nAvailable Currencies:\n";

12 cout << "1. INR\n";

13 cout << "2. USD\n";

14 cout << "3. EUR\n";

15 cout << "4. GBP\n";

16 cout << "5. JPY\n";

17 cout << "6. AUD\n";

18 }

19

20 // Function to get currency name

21 string getCurrencyName(int choice) {

22 if(choice == 1) return "INR";

23 if(choice == 2) return "USD";

24 if(choice == 3) return "EUR";

25 if(choice == 4) return "GBP";

26 if(choice == 5) return "JPY";

27 if(choice == 6) return "AUD";

28 return "Unknown";

29 }

30

31 // Exchange rates (basic fixed rates)

32 float getRate(int from, int to) {

33

34 if (from == to) return 1;

35

36 // Base INR rates

Ln 149, Col 2

Spaces: 4

UTF-8

LF

C++

Go Live

Mac

EXPLORER

OPEN EDITORS

Welcome

main.cpp

CURRENCY EXCHANGE & ...

main

main.cpp

transactions.txt

OUTLINE

TIMELINE

Welcome

main.cpp

convertCurrency()

32 float getRate(int from, int to) {

35

36 // Base INR rates

37 if (from == 1 && to == 2) return 0.012;

38 if (from == 1 && to == 3) return 0.011;

39 if (from == 1 && to == 4) return 0.0095;

40 if (from == 1 && to == 5) return 1.65;

41 if (from == 1 && to == 6) return 0.018;

42

43 if (from == 2 && to == 1) return 83;

44 if (from == 3 && to == 1) return 90;

45 if (from == 4 && to == 1) return 105;

46 if (from == 5 && to == 1) return 0.60;

47 if (from == 6 && to == 1) return 55;

48

49 return 1; // simplified for demo

50 }

51

52 // Function to get current date and time

53 string getDateTime() {

54 time_t now = time(0);

55 char* dt = ctime(&now);

56 return string(dt);

57 }

58

59 // Convert currency

60 void convertCurrency() {

61 int from, to;

62 float amount;

63

64 showCurrencies();

65 cout << "\nEnter From Currency (1-6): ";

66 cin >> from;

67

68 cout << "Enter To Currency (1-6): ";

69 cin >> to;

Ln 65, Col 45

Spaces: 4

UTF-8

LF

C++

Go Live

Mac

EXPLOREDOR

OPEN EDITORS

main.cpp

CURRENCY EXCHANGE & ...

main

main.cpp

transactions.txt

OUTLINE

TIMELINE

Welcome

main.cpp

60 void convertCurrency() {
71 cout << "Enter Amount: ";
72 cin >> amount;
73
74 float rate = getRate(from, to);
75 float result = amount * rate;
76
77 cout << fixed << setprecision(2);
78 cout << "\nConverted Amount: " << result
79 << " " << getCurrencyName(to) << endl;
80
81 ofstream file("transactions.txt", ios::app);
82
83 file << "\n-----\n";
84 file << "Date & Time: " << getDateTine();
85 file << "From: " << getCurrencyName(from) << endl;
86 file << "To: " << getCurrencyName(to) << endl;
87 file << "Amount: " << amount << endl;
88 file << "Converted Amount: " << result << endl;
89
90 file.close();
91
92 transactionCount++;
93
94 cout << "\nTransaction Saved Successfully!\n";
95
96
97 // View transaction history
98 void viewTransactions() {
99 ifstream file("transactions.txt");
100 string line;
101
102 cout << "\n----- Transaction History ----- \n";
103
104 while(getline(file, line)) {
105 cout << line << endl;
106
107 }
108 }
109 }
110 }

Welcome to Copilot

Let's get started

Add context (#), external files, or a workspace

Build Workspace

Show Config

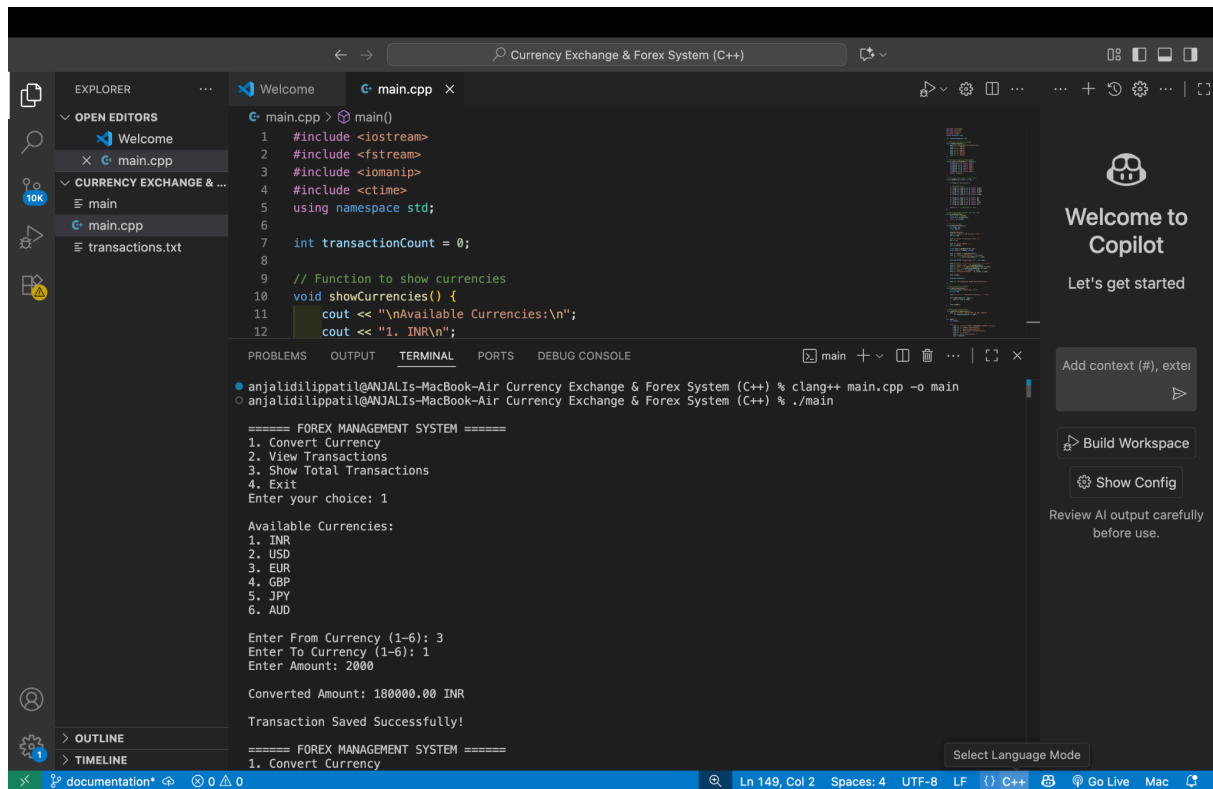
Review AI output carefully before use.

Ln 65, Col 45 Spaces: 4 UTF-8 LF C++ Go Live Mac

```
98 void viewTransactions() {
104     while(getline(file, line)) {
105         cout << line << endl;
106     }
107     file.close();
108 }
109
110 // Show total transactions
111 void showTotalTransactions() {
112     cout << "\nTotal Transactions in this session: "
113     << transactionCount << endl;
114 }
115
116 int main() {
117     int choice;
118
119     do {
120         cout << "\n===== FOREX MANAGEMENT SYSTEM =====\n";
121         cout << "1. Convert Currency\n";
122         cout << "2. View Transactions\n";
123         cout << "3. Show Total Transactions\n";
124         cout << "4. Exit\n";
125         cout << "Enter your choice: ";
126         cin >> choice;
127
128         switch(choice) {
129             case 1:
130                 convertCurrency();
131                 break;
132             case 2:
133                 viewTransactions();
134                 break;
135             case 3:
136                 showTotalTransactions();
137                 break;
138         }
139     } while(choice != 4);
140
141     return 0;
142 }
```

```
117 int main() {
120     do {
128
129         switch(choice) {
130             case 1:
131                 convertCurrency();
132                 break;
133             case 2:
134                 viewTransactions();
135                 break;
136             case 3:
137                 showTotalTransactions();
138                 break;
139             case 4:
140                 cout << "\nThank you for using Forex System!\n";
141                 break;
142             default:
143                 cout << "\nInvalid Choice! Try Again.\n";
144         }
145     } while(choice != 4);
146
147     return 0;
148 }
149 }
```


OUTPUT



```
main.cpp
1 #include <iostream>
2 #include <fstream>
3 #include <iomanip>
4 #include <ctime>
5 using namespace std;
6
7 int transactionCount = 0;
8
9 // Function to show currencies
10 void showCurrencies() {
11     cout << "\nAvailable Currencies:\n";
12     cout << "1. INR\n";
13 }
14
15 int main() {
16     showCurrencies();
17     int choice;
18     while (choice != 4) {
19         cout << "\n===== FOREX MANAGEMENT SYSTEM =====\n";
20         cout << "1. Convert Currency\n";
21         cout << "2. View Transactions\n";
22         cout << "3. Show Total Transactions\n";
23         cout << "4. Exit\n";
24         cout << "Enter your choice: ";
25         int choice;
26         while (choice < 1 || choice > 4) {
27             choice = 0;
28             while (choice < 1 || choice > 4) {
29                 choice = 0;
30             }
31             cout << "Invalid choice. Please enter a valid choice (1-4): ";
32             choice = 0;
33         }
34         if (choice == 1) {
35             int fromCurrency, toCurrency;
36             while (fromCurrency < 1 || fromCurrency > 6 || toCurrency < 1 || toCurrency > 6) {
37                 cout << "Enter From Currency (1-6): ";
38                 fromCurrency = 0;
39                 while (fromCurrency < 1 || fromCurrency > 6) {
40                     fromCurrency = 0;
41                 }
42                 cout << "Enter To Currency (1-6): ";
43                 toCurrency = 0;
44                 while (toCurrency < 1 || toCurrency > 6) {
45                     toCurrency = 0;
46                 }
47             }
48             double amount;
49             while (amount < 0) {
50                 cout << "Enter Amount: ";
51                 amount = 0;
52             }
53             double convertedAmount;
54             if (fromCurrency == toCurrency) {
55                 convertedAmount = amount;
56             } else {
57                 convertedAmount = amount * 180000.00 / 1000.00;
58             }
59             cout << "Converted Amount: " << convertedAmount << " INR\n";
60             cout << "Transaction Saved Successfully!\n";
61             transactionCount++;
62             fstream file("transactions.txt", ios::app);
63             if (!file.is_open()) {
64                 cout << "Error opening file!\n";
65                 return 1;
66             }
67             file << "Transaction " << transactionCount << ": " << fromCurrency << " to " << toCurrency << " INR, Amount: " << amount << " INR, Converted Amount: " << convertedAmount << " INR\n";
68             file.close();
69         } else if (choice == 2) {
70             cout << "Transaction History\n";
71             fstream file("transactions.txt", ios::in);
72             if (!file.is_open()) {
73                 cout << "Error opening file!\n";
74                 return 1;
75             }
76             while (!file.eof()) {
77                 string line;
78                 getline(file, line);
79                 cout << line << "\n";
80             }
81             file.close();
82         } else if (choice == 3) {
83             cout << "Total Transactions in this session: " << transactionCount << "\n";
84         }
85     }
86     return 0;
87 }
```

```
anjalidilipatil@ANJALIS-MacBook-Air Currency Exchange & Forex System (C++) % clang++ main.cpp -o main
anjalidilipatil@ANJALIS-MacBook-Air Currency Exchange & Forex System (C++) % ./main

===== FOREX MANAGEMENT SYSTEM =====
1. Convert Currency
2. View Transactions
3. Show Total Transactions
4. Exit
Enter your choice: 1

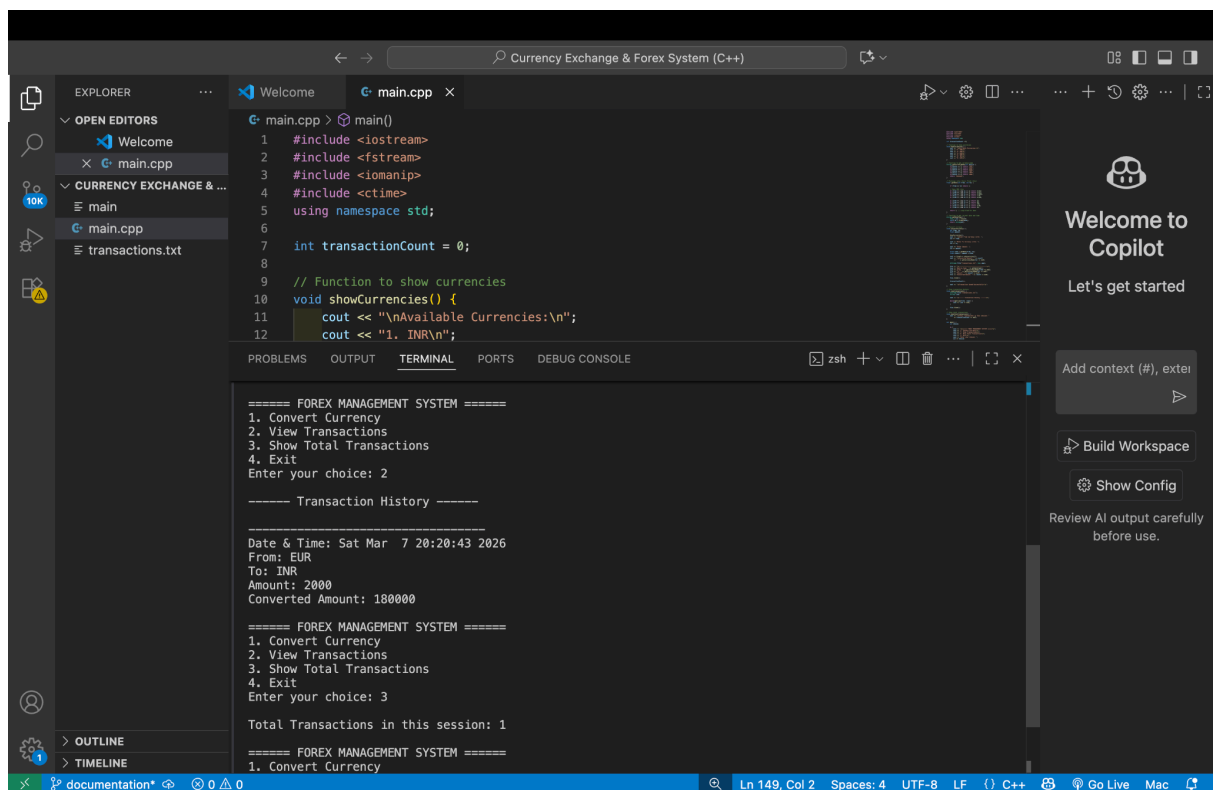
Available Currencies:
1. INR
2. USD
3. EUR
4. GBP
5. JPY
6. AUD

Enter From Currency (1-6): 3
Enter To Currency (1-6): 1
Enter Amount: 2000

Converted Amount: 180000.00 INR

Transaction Saved Successfully!

===== FOREX MANAGEMENT SYSTEM =====
1. Convert Currency
```



```
main.cpp
1 #include <iostream>
2 #include <fstream>
3 #include <iomanip>
4 #include <ctime>
5 using namespace std;
6
7 int transactionCount = 0;
8
9 // Function to show currencies
10 void showCurrencies() {
11     cout << "\nAvailable Currencies:\n";
12     cout << "1. INR\n";
13 }
14
15 int main() {
16     showCurrencies();
17     int choice;
18     while (choice != 4) {
19         cout << "\n===== FOREX MANAGEMENT SYSTEM =====\n";
20         cout << "1. Convert Currency\n";
21         cout << "2. View Transactions\n";
22         cout << "3. Show Total Transactions\n";
23         cout << "4. Exit\n";
24         cout << "Enter your choice: ";
25         int choice;
26         while (choice < 1 || choice > 4) {
27             choice = 0;
28             while (choice < 1 || choice > 4) {
29                 choice = 0;
30             }
31             cout << "Invalid choice. Please enter a valid choice (1-4): ";
32             choice = 0;
33         }
34         if (choice == 1) {
35             int fromCurrency, toCurrency;
36             while (fromCurrency < 1 || fromCurrency > 6 || toCurrency < 1 || toCurrency > 6) {
37                 cout << "Enter From Currency (1-6): ";
38                 fromCurrency = 0;
39                 while (fromCurrency < 1 || fromCurrency > 6) {
40                     fromCurrency = 0;
41                 }
34                 cout << "Enter To Currency (1-6): ";
35                 toCurrency = 0;
36                 while (toCurrency < 1 || toCurrency > 6) {
37                     toCurrency = 0;
38                 }
39             }
40             double amount;
41             while (amount < 0) {
42                 cout << "Enter Amount: ";
43                 amount = 0;
44             }
45             double convertedAmount;
46             if (fromCurrency == toCurrency) {
47                 convertedAmount = amount;
48             } else {
49                 convertedAmount = amount * 180000.00 / 1000.00;
50             }
51             cout << "Converted Amount: " << convertedAmount << " INR\n";
52             cout << "Transaction Saved Successfully!\n";
53             transactionCount++;
54             fstream file("transactions.txt", ios::app);
55             if (!file.is_open()) {
56                 cout << "Error opening file!\n";
57                 return 1;
58             }
59             file << "Transaction " << transactionCount << ": " << fromCurrency << " to " << toCurrency << " INR, Amount: " << amount << " INR, Converted Amount: " << convertedAmount << " INR\n";
60             file.close();
61         } else if (choice == 2) {
62             cout << "Transaction History\n";
63             fstream file("transactions.txt", ios::in);
64             if (!file.is_open()) {
65                 cout << "Error opening file!\n";
66                 return 1;
67             }
68             while (!file.eof()) {
69                 string line;
70                 getline(file, line);
71                 cout << line << "\n";
72             }
73             file.close();
74         } else if (choice == 3) {
75             cout << "Total Transactions in this session: " << transactionCount << "\n";
76         }
77     }
78     return 0;
79 }
```

```
===== FOREX MANAGEMENT SYSTEM =====
1. Convert Currency
2. View Transactions
3. Show Total Transactions
4. Exit
Enter your choice: 2

----- Transaction History -----

Date & Time: Sat Mar 7 20:20:43 2026
From: EUR
To: INR
Amount: 2000
Converted Amount: 180000

===== FOREX MANAGEMENT SYSTEM =====
1. Convert Currency
2. View Transactions
3. Show Total Transactions
4. Exit
Enter your choice: 3

Total Transactions in this session: 1

===== FOREX MANAGEMENT SYSTEM =====
1. Convert Currency
```

6. Problem Statement / Case Study Results and Conclusion.

Results

The Forex Management System successfully performs currency conversions between different currencies. The program allows users to enter an amount, select currencies, and view the converted value instantly. The system also records each transaction in a file along with date and time.

Conclusion

This project demonstrates how C++ can be used to create a simple financial application. The Forex Management System improves accuracy and efficiency in currency conversion. The system can be further improved by adding real-time exchange rates, graphical user interfaces, and additional currencies.

7. References

C++ Programming Tutorials – W3Schools

LearnCPP Programming Guide

C++ Course Notes and Class Materials

Documentation of C++ Standard Library